



$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?

• Part 2 (Rakotomalala & W, in review, 2026):

<sup>o</sup> **Theorem.** There is a universal  $c > 0$  such that if  $\sigma_x < cPe/2\pi$  and  $\sigma_\theta$  small enough, then there is a value of  $\chi$  at which there exist two branches representing two families of stationary solutions: one constant in one spatial  $x$ -variable (lanes) and one swap-symmetric:  
 $f(x_1, x_2, \theta) = f(-x_2, -x_1, -\theta - \pi/2)$  (spots)

◦ Existence of critical  $\tau = \tau^*$ 's with super- to subcritical transitions



For lanes and spots separately and  $0 < \tau_s^* < \tau_l^*$

**Corollary.** For  $\tau < \tau_s^*$  spots are locally nonlinearly stable and lanes unstable (up to translation)



**Dynamics**

# Dynamics

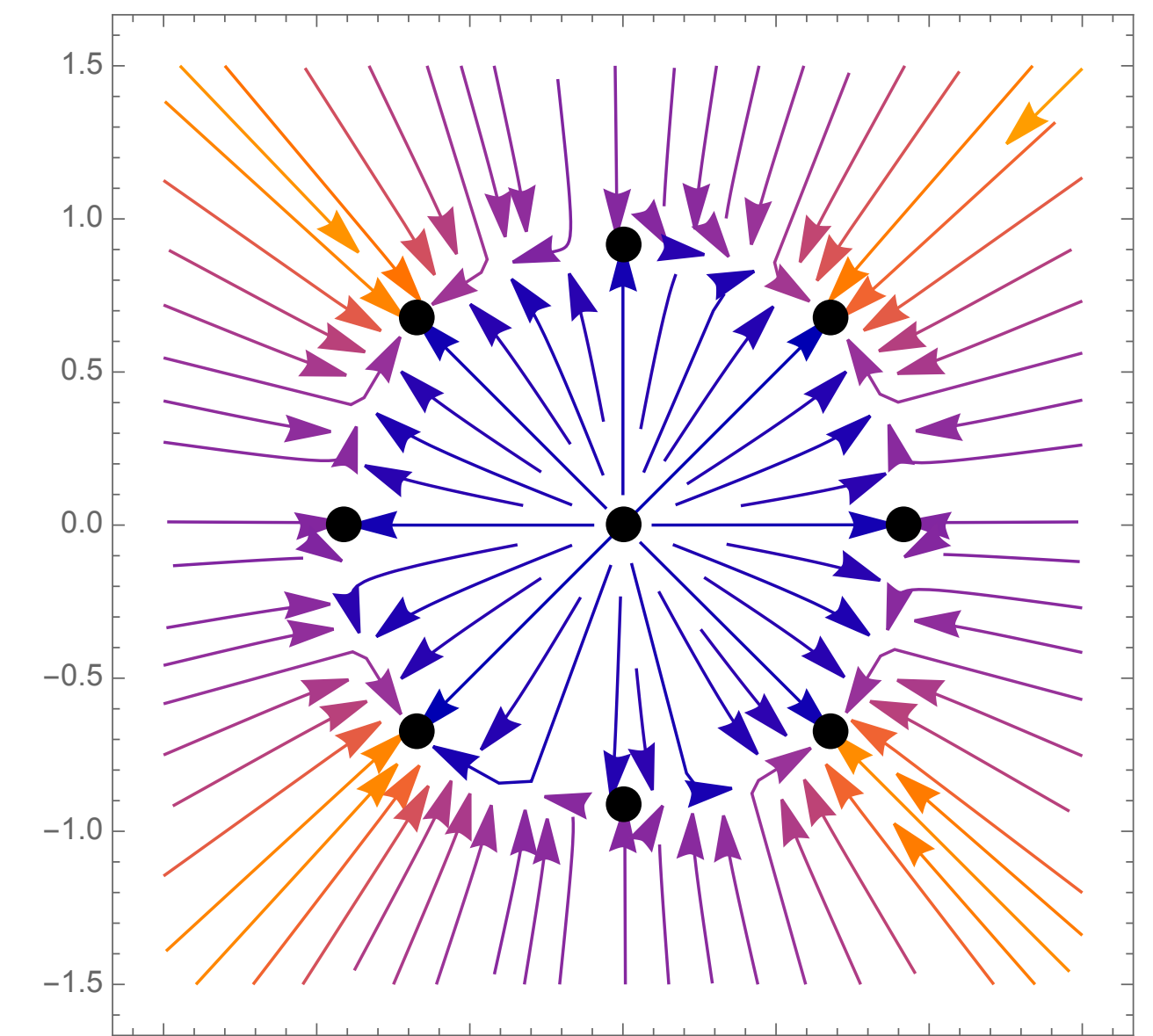
$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta)f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta))f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?
- Part 2 (Rakotomalala & dW, in review, 2026):
  - **Theorem.** There is a universal  $c > 0$  such that if  $\sigma_x < c\text{Pe}/2\pi$  and  $\sigma_\theta$  small enough, then there is a value of  $\chi$  at which there exist two branches representing two families of stationary solutions: one constant in one spatial  $x$ -variable (lanes) and one swap-symmetric:  $f(x_1, x_2, \theta) = f(-x_2, -x_1, -\theta - \pi/2)$  (spots)
  - Existence of critical  $\tau = \tau^*$ 's with super- to subcritical transitions
    - For lanes and spots separately and  $0 < \tau_s^* < \tau_l^*$
  - **Corollary.** For  $\tau < \tau_s^*$  spots are locally nonlinearly stable and lanes unstable (up to translation)

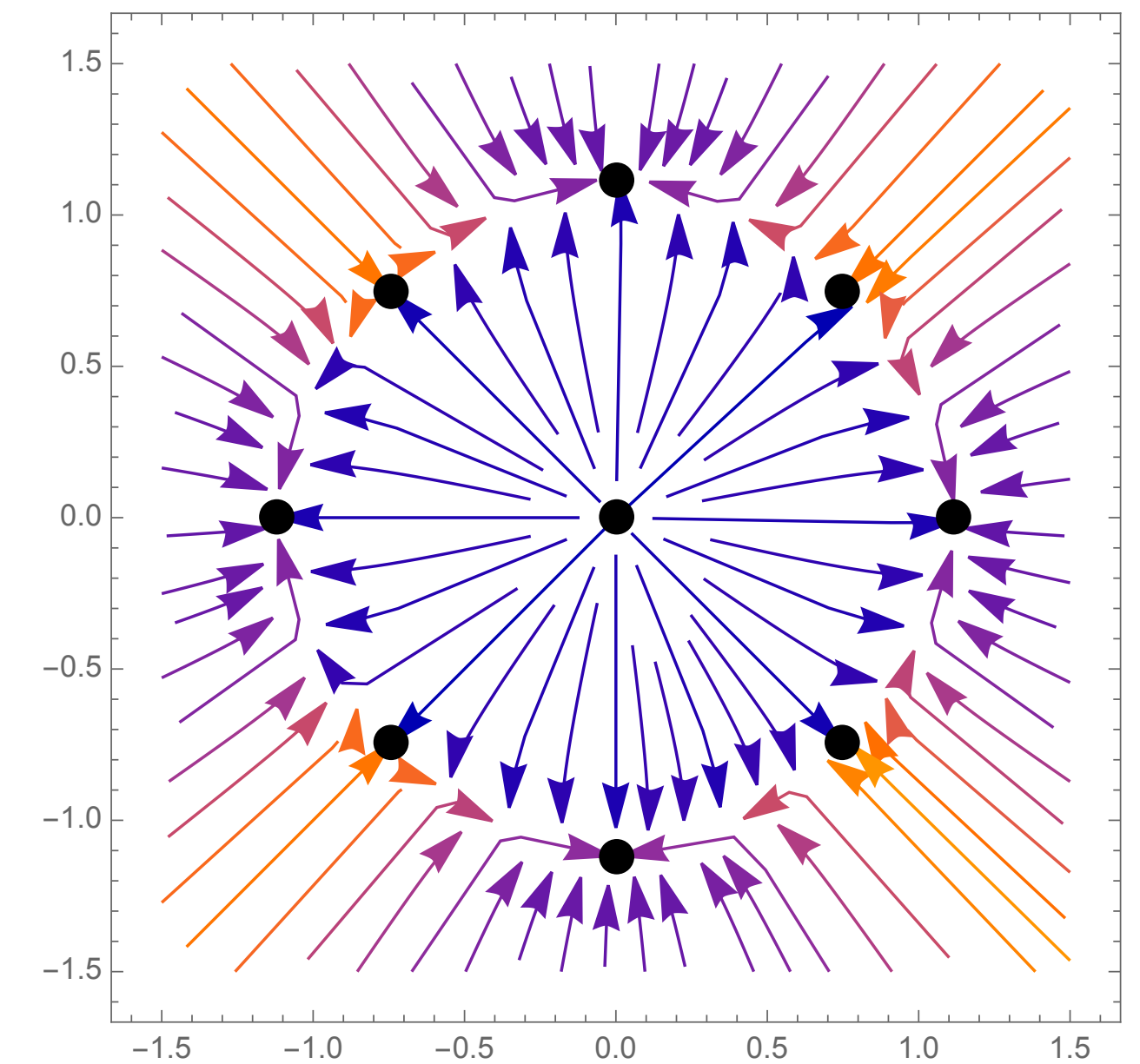
# Dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) \\ &+ \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as  $t \rightarrow \infty$ ?
- Part 2 (Rakotomalala & dW, in review, 2026):
  - Existence of two families of non-trivial stationary states



$$\tau < \tau^*$$



$$\tau > \tau^*$$